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Pharmacokinetics

Next time you are reviewing the chart for your patient, take a look at the medication administration record. Every entry represents a decision – a decision to alter the physiology of your patient, with all the beneficial and adverse consequences that go along with that decision. Medication administration while on extracorporeal membrane oxygenation (ECMO) takes the weight of these decisions to another level, adding the uncertainty of how that medication will interact with the circuit itself.

This chapter will review principles to consider when making the decision to initiate, continue, or remove a medication for a patient on ECMO. First, let's consider a couple of important caveats.

There is a relative paucity of data regarding the pharmacokinetics of drugs for patients on ECMO. How a drug interacts with the circuit, with other drugs, and with the body in the setting of severe critical illness is not completely understood, and the uniqueness of these interactions extends to all patients. Every patient may react to medications differently, and the impact of the circuit on medication effect may differ from patient to patient.

This chapter will try to introduce some basic principles that can be the framework for approaching medication administration for patients on ECMO, and how to assess the effect of these medications.

NORMAL EFFECT OF A DRUG ON THE BODY

Every time we hang an IV medication, initiate a drip, or administer a pill, there are two interactions occurring, how the drug interacts with the body and how the body interacts with the drug. Let's consider the ways the body interacts with the drug, referred to as pharmacokinetics (Fig. 21.1).

1. **Absorption:** how the drug enters the body, determined by route, solubility, and presence of pumps/receptors
2. **Distribution:** how the drug is disseminated to the target cells, determined by volume of distribution and how it is bound

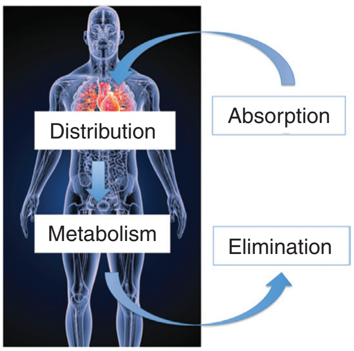

FIG. 21.1 The mechanisms by which the body interacts with a drug. (Modified from Sciencepics/Shutterstock.com.)

3. **Metabolism:** the conversion of drug to enhance removal
4. **Elimination:** the removal of metabolized drug outside the body

These processes allow for a pattern of how much drug is present in the body based on when it is administered. What does this pattern look like? Let's consider what happens when we administer a medication to a patient.

The first thing that happens is an increase in the amount of drug that is available to be distributed. How fast this increase happens is determined by the adsorption and distribution of the drug. Regardless, you can imagine a pattern like this, with a sharp uptick in drug available and an eventual tapering off toward the top of the curve (Fig. 21.2).

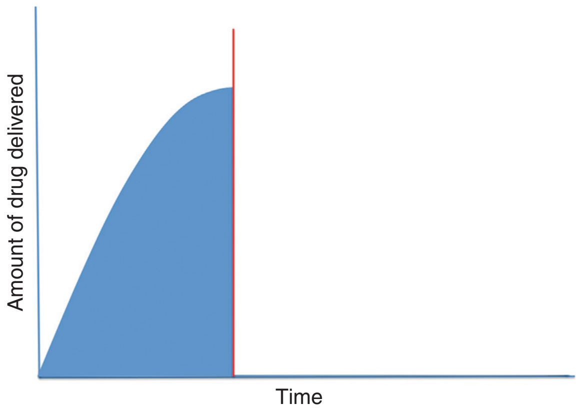

FIG. 21.2 Drug entrance into the body through absorbtion and distribution.

As time goes on, there is less drug available for distribution and more drug begins to be metabolized and eliminated. Initially, this leads to a tapering off of the increase, and then leads to a decrease in drug available, starting slowly at first and then becoming more rapid as time goes on, giving rise to a bell curve shape.

The area under this curve represents the total drug exposure, which leads to the overall effect of the drug. This is important, as it explains the different ways that drugs can achieve their target effect – a drug that is quickly absorbed and slowly eliminated may have a very different overall effect than a drug that is slowly absorbed and quickly eliminated (Fig. 21.3).

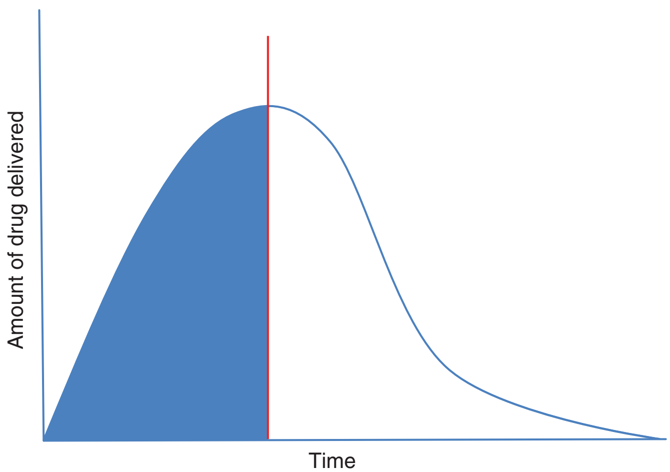

FIG. 21.3 Drug entering the body, reaching a peak level (red line), and eventual removal

How quickly the drug is absorbed and eliminated determines the height of the bell curve, while the distribution and metabolism of the drug impacts the width of the bell curve. Distribution describes how widely the drug is dispersed throughout the body, encompassing two characteristics: volume of distribution (Vd) and binding (Fig. 21.4).

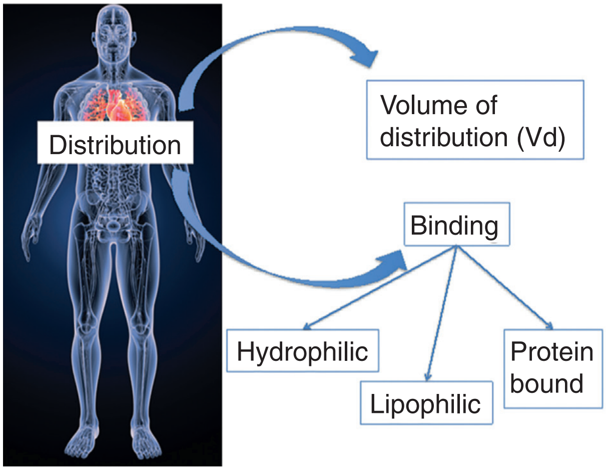

FIG. 21.4 Distribution as a function of volume of distribution and binding. (Modified from Sciencepics/Shutterstock.com.)

Vd describes the relative fraction of drug in the blood versus the extravascular space. Drug that absorbs into the fat, muscle, or interstitium has a much wider volume of distribution, with a consequent broadening effect on the bell curve.

How the drug is bound can be classified as hydrophilic, lipophilic, and protein bound. Hydrophilic drugs have a lower Vd and are more dependent on fluid shifts, while lipophilic drugs generally have the ability to penetrate into tissues with a consequent lower Vd. Protein binding generally restricts Vd as bound drugs are less likely to cross membranes.

However, we must also put this in the context of the amount of drug that is needed to achieve the clinical effect (**minimal effective concentration**) and the amount of drug that can have an adverse effect (**toxic concentration**) (Fig. 21.5).

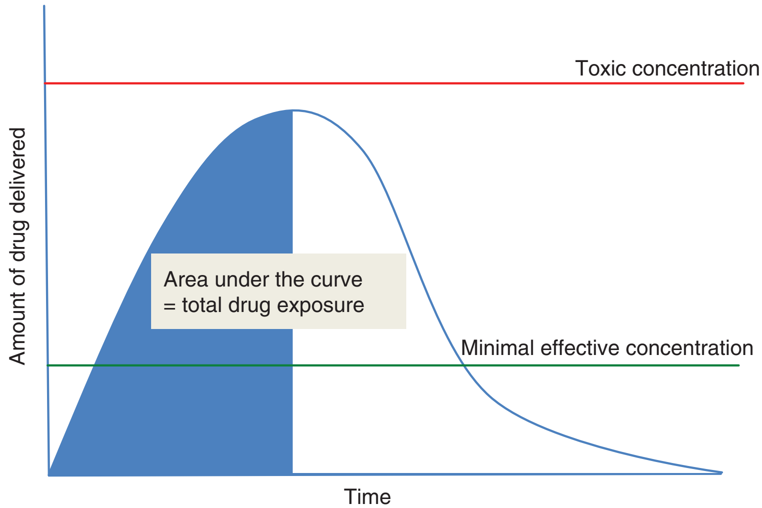

FIG. 21.5 Total drug exposure as a function of minimal effective concentration and toxic concentration.

The key to obtaining maximal therapeutic effectiveness of any drug is to have the maximum area under the curve above the minimal effective concentration while remaining under the toxic concentration as much as possible.

VARIATIONS IN PHARMACOKINETICS: EFFECTS OF CRITICAL ILLNESS

Taken together, this can be a very dynamic process that may differ from patient to patient and even day-to-day based on the clinical circumstances. How can this change? Let's consider just a few ways these processes can be altered by critical illness alone.

Absorption can be decreased in the setting of shock due to decreased gastrointestinal and subcutaneous perfusion. **Distribution** can be altered due to decreased albumin and poor end organ perfusion. **Metabolism** can be altered by reduced hepatic blood flow and altered hepatic enzyme function. **Elimination** of drugs can be altered by acute kidney injury and altered active transport of medications.

The overall effect of critical illness can result in higher or lower drug exposure as illustrated in Fig. 21.6. Consider the effect of a patient in sepsis with septic shock. This patient could have a high cardiac output due to catecholamine surge/vasodilation shock, leaky capillaries due to endotoxin effect, and altered protein binding due to acidosis, all leading to decreased drug exposure.

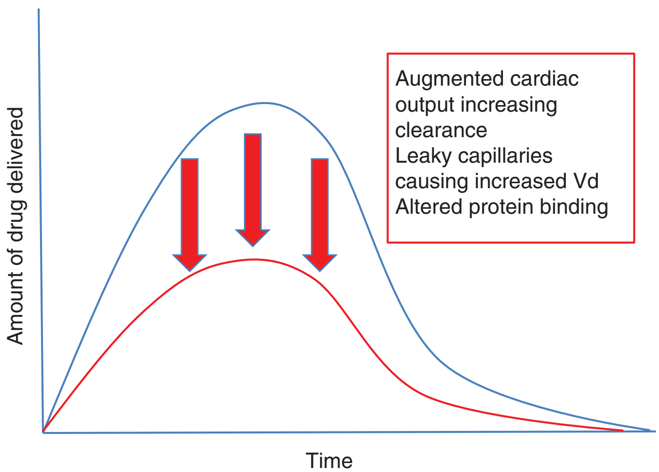

FIG. 21.6 Potential mechanisms for lower drug exposure in critical illness.

On the other hand, the patient may have decreased protein binding due to low albumin, altered metabolism due to hepatic dysfunction, and impaired elimination due to acute kidney injury, all of which could possibly lead to higher than expected drug exposure as illustrated in Fig. 21.7.

To this point, we have hopefully established that pharmacokinetics is a dynamic process that can be impacted by characteristics unique to the drug itself, to the patient, or to the overall state of critical illness. This is not to say that we have no idea what is going to happen when we administer a medication, but rather that the response can be varied. This varied response can impact the therapeutic/toxic effect of the medication, and we should maintain vigilance for monitoring both the therapeutic and toxic effects of these medications.

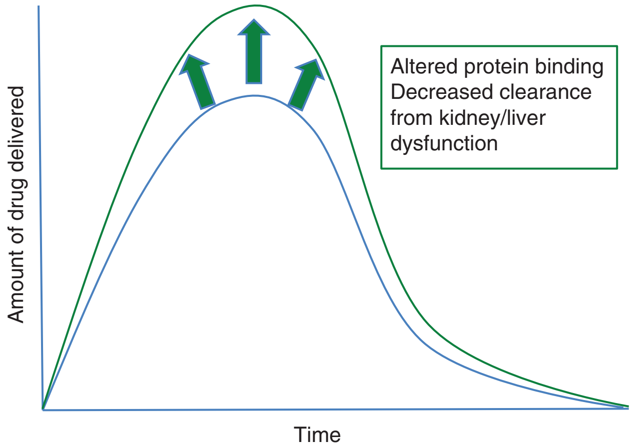

FIG. 21.7 Potential mechanisms for higher drug exposure in critical illness.

CHANGES TO PHARMACOKINETICS DURING ECMO

Let's now take our discussion one step further, considering the changes to how drug is absorbed, distributed, metabolized, and eliminated for patients on ECMO. Besides the effect of critical illness, consider two major differences which must be accounted for in the patient on extracorporeal support: effect of the circuit/tubing and effect of the oxygenator.

1. **Effect of circuit/tubing:** Remember when we initiate ECMO, we connect to a circuit primed with 500–600 mL of crystalloid. This effectively increases the Vd by the priming volume to the original Vd and has the additional effect of fluid shift/hemodilution associated with a 500–600 mL fluid bolus (Fig. 21.8).
2. **Effect of the oxygenator:** The oxygenator is made of plastic tubules and foreign substances that come into contact with blood. Based on the protein binding/lipophilicity/hydrophilicity of the drug, the oxygenator may augment clearance of the drug, or preferentially bind to the drug, sequestering it in the membrane (Fig. 21.8).

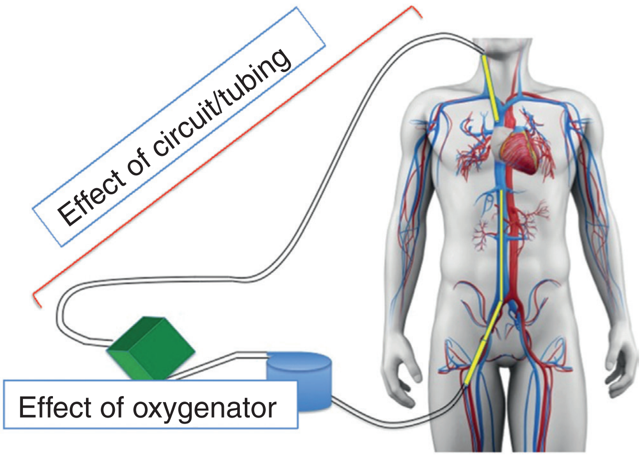

FIG. 21.8 Pharmacokinetics on ECMO: the effect of the circuit/tubing and the effect of the oxygenator. (Modified from SciePro/Shutterstock.com.)

While the effect of augmented Vd is relatively easy to account for, the effect of the oxygenator can be more varied, dependent on the age and function of the oxygenator and other host factors. As a general rule, **hydrophilic drugs** are more likely to be affected by priming/hemodilution, **lipophilic drugs** are more likely to be sequestered by the oxygenator membrane, and **protein bound drugs** lead to greater clearance by circuit.

Although data are scarce for many drugs, there are some principles that we will describe which can hopefully be applied clinically at the bedside.

ECMO AND SEDATION: THE EFFECTS OF PROTEIN BINDING

Clearance of commonly used sedatives is a good example of the role of protein binding and its role in drug clearance by the membrane oxygenator. Most IV sedation/analgesic medications that are commonly used in the ICU such as propofol, dexmedetomidine, fentanyl, lorazepam, and midazolam are highly protein bound, with protein binding occurring in approximately 80%–90% of drug concentration in the blood.

The result is a much steeper bell curve, with a consequent lower area under the curve and less overall drug exposure, as illustrated in Fig. 21.9

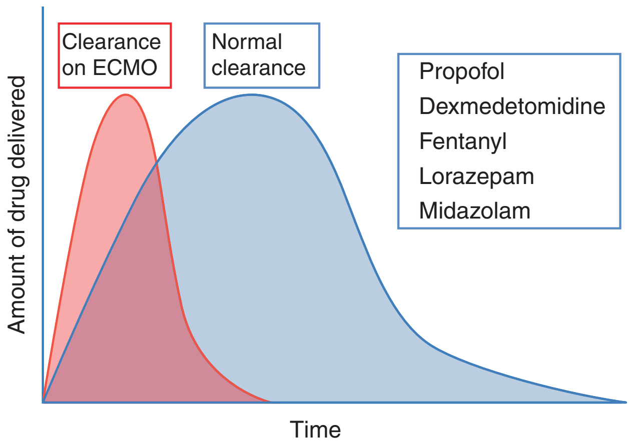

FIG. 21.9 The role of protein binding on increased drug clearance for a patient on ECMO.

Contrast this to morphine and morphine derivatives, such as hydromorphone. These sedative medications by contrast are only 10%–20% protein bound in the blood, leading to a much smaller overall effect of the oxygenator on drug clearance (Fig. 21.10).

What is the clinical result of these differences in protein binding/drug clearance from the oxygenator?

You may observe a pattern as illustrated in Fig. 21.11, which is adapted from a trial where these drugs were administered to a running circuit and the levels were measured over the course of 24 hours. While the relative concentration of morphine remained essentially unchanged, the concentration of midazolam, fentanyl, and propofol fell precipitously.

Does this mean that you should use one medication over another while on ECMO? Not necessarily. Each drug has its own profile of intended and adverse effects and needs to be tailored to the clinical endpoint. Rather, this example is to illustrate the clinical effects of the oxygenator, which may be very different than a patient who is not on ECMO. For teams still wanting to use a drug with high clearance, dose adjustment may be required with careful observation for therapeutic and toxic effects at the higher dosage.

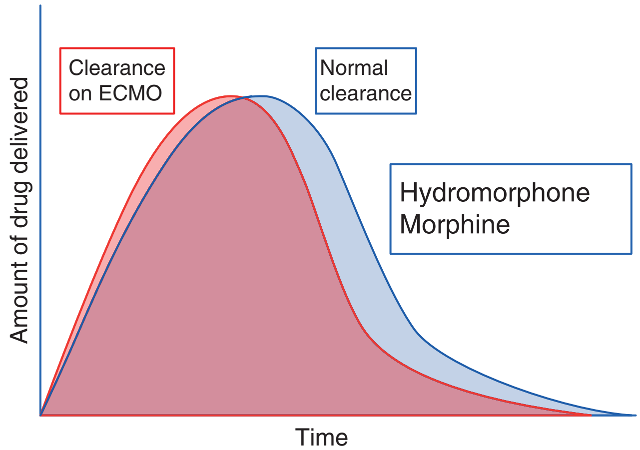

FIG. 21.10 Lower circuit clearance of drugs with lower protein binding.

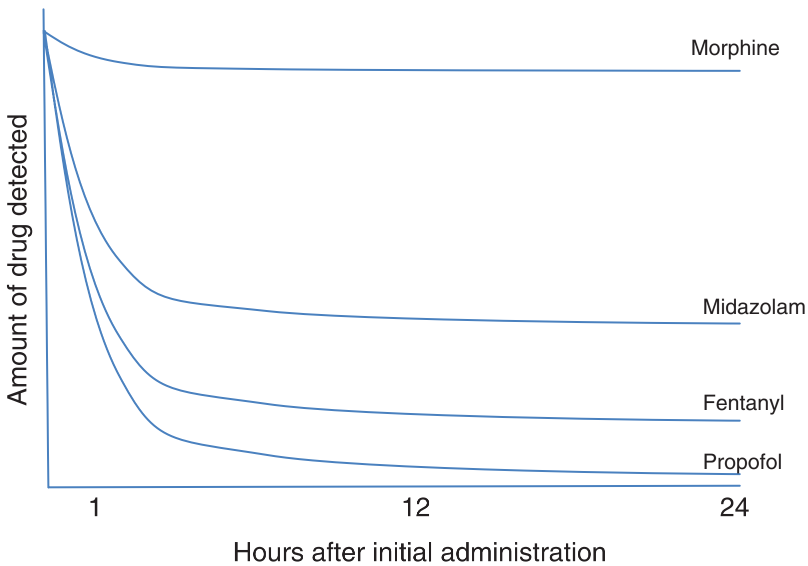

FIG. 21.11 Schematic of drug clearance rates for different sedation medications.

PHARMACOKINETICS OF ANTIBIOTICS ON ECMO

Antibiotics are another example where pharmacokinetics on ECMO can be unpredictable or at the very least, different from patients not on ECMO support. In the case of antibiotics, the therapeutic window can be narrower, with a smaller gap between the minimum effective concentration and the toxic concentration in many cases (Fig. 21.12).

As an example, beta lactams are often hydrophilic, meaning that the higher Vd can decrease the area under the curve that occurs above the minimum effective concentration. The effect of this lower drug exposure may mean less effective activity against pathogens, particularly virulent versions requiring higher concentrations of drug.

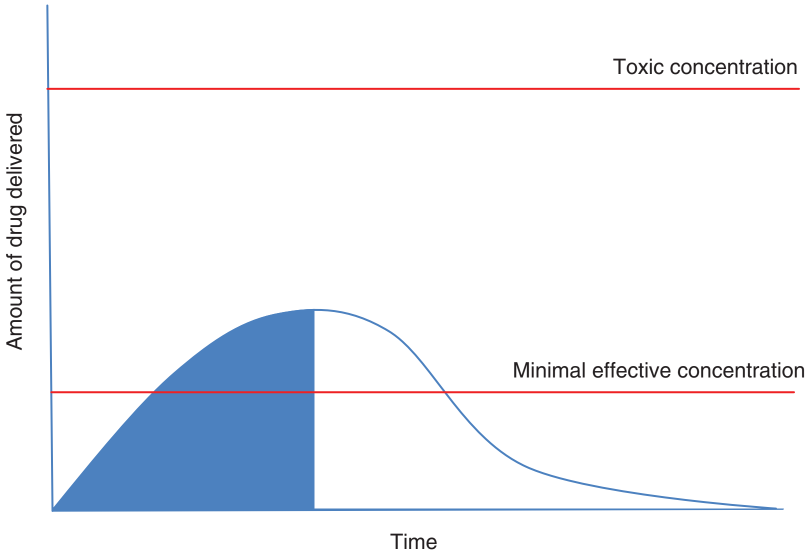

FIG. 21.12 Area under the curve above the minimal effective concentration for antibiotics.

Other drugs may undergo clearance by the oxygenator, similar to the way described for sedation. Although this clearance is not usually as profound as is observed for sedatives, it does exist due to relative differences in protein binding of different antibiotics and should be considered when selecting/dosing antibiotics (Fig. 21.13).

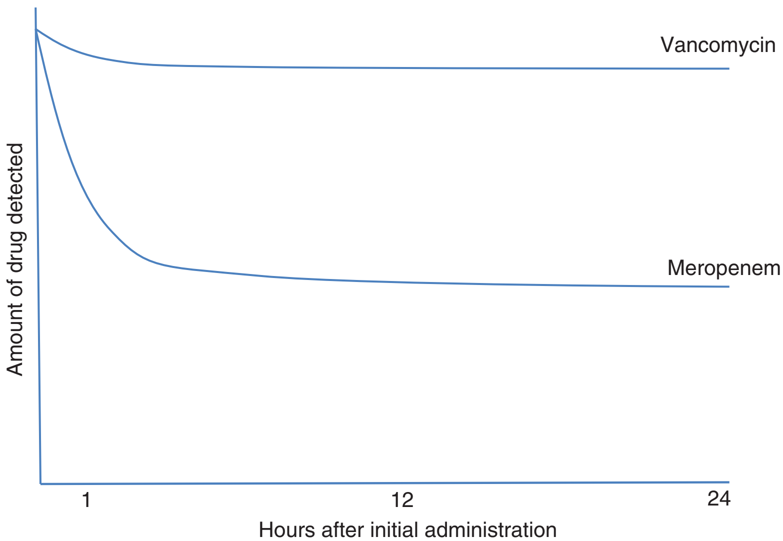

FIG. 21.13 Relative clearance rates of antibiotics by ECMO circuit.

The consequence of these phenomena may mean adjustments to how antibiotics are dosed, timed, or administered. Examples of these adjustments include extended administration time, as illustrated in Fig. 21.14, which increases the area under the curve above the minimum effective concentration while also avoiding toxic concentrations. Other strategies are higher dosing, more frequent dosing, or adding a loading dose.

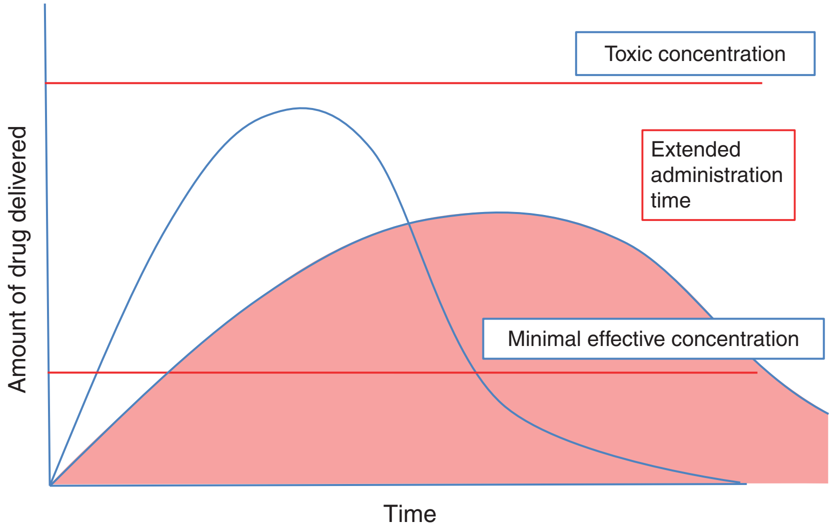

FIG. 21.14 Extended administration time as a strategy for increasing the area under the curve for antibiotic administration.

The other strategy to inform dosage of antibiotics and other medications in the setting of unpredictable pharmacokinetics of ECMO is to measure levels directly. These levels can be correlated to toxic/therapeutic concentrations of the medications, with dose adjustment as needed.

PUTTING IT TOGETHER

Alterations in pharmacokinetics are going to exist – it is simply the nature of the dynamic way that our bodies absorb, distribute, metabolize, and eliminate drugs. Unfortunately, the unpredictability of these pharmacokinetics often increases in ECMO due to the changes associated with critical illness as well as the effect ECMO circuit/oxygenator.

The temptation is to think of the ECMO circuit as a black box and conclude that the true effect cannot be known. However, we can easily identify and should carefully consider the therapeutic and toxic effects of every medication we administer. Whenever possible, measure and monitor drug levels, optimizing therapeutic effect with dosing adjustments as necessary and with the consultation of the pharmacy team. Doing so will help to illustrate the way that these drugs are interacting with the body in the way that is most meaningful to the patient, by maximizing benefit and minimizing harm.

We have advocated for this strategy many times throughout this book, but let's revisit one final time in the context of pharmacokinetics. Effective clinical decision-making should be informed by making a prediction of the intended effect/harm of a medication, observing for that effect, and then adjusting course as needed thereafter. This measured approach can allow for an optimized use of medications every time they are administered to our patients.

SUGGESTED READING

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